ICM we just built, except they used a more sophisticated actor-critic model called *proximal policy optimization* (PPO) rather than DQN.

An experiment you can try is to replace the encoder network with a *random projection*. A random projection just means multiplying the input data by a randomly initialized matrix (e.g., a randomly initialized neural network that is fixed and not trained). The Burda et al. 2018 paper demonstrated that a random projection works almost as well as the trained encoder.

8.7. Alternative intrinsic reward mechanisms

In this chapter we described the serious problem faced by RL agents in environments with sparse rewards. We considered the solution to be imbuing agents with a sense of curiosity, and we implemented an approach from the Pathak et al. 2017 paper, one of the most widely cited papers in reinforcement learning research in recent years. We chose to demonstrate this approach not just because it is popular, but because it builds on what we've learned in previous chapters without introducing too many new notions. Curiosity-based learning (which goes by many names) is a very active area of research, and there are many alternative approaches, some of which we think are better than the ICM.

Many of the other exciting methods use Bayesian inference and information theory to come up with novel mechanisms to drive curiosity. The prediction error (PE) approach we used in this chapter is just one implementation under a broader PE umbrella. The basic idea, as you now know, is that the agent wants to reduce its PE (or in other words, its uncertainty about the environment), but it must do so by actively seeking out novelty lest it be surprised by something unexpected.

Another umbrella is that of agent *empowerment*. Rather than seeking to minimize prediction error and make the environment more predictable, empowerment strategies optimize the agent to maximize its control over the environment (figure 8.21). One paper in this area is "Variational Information Maximisation for Intrinsically Motivated Reinforcement